PRIYANKA MODI
24BCE0813

CAT-1 (D1+TD1) Winter Sem.

Q1. (a) i) error correct: MUL AB
→ comma not used

(ii) error : PUSH ACC
→ accept only direct address

(iii) error
→ destination must be accumulator or direct address
: ANL A, R0

(iv) error
→ R2–R7 cannot be used as pointers, only R_0 & R_1 supports indirect addressing.
: MOV A, @R0

(v) error
→ Register to register moves not allowed
: MOV A, R1
MOV R2, A

(b) (i) Direct addressing mode
(ii) Immediate addressing mode
(iii) Register addressing
(iv) Indexed addressing
(v) Register Indirect addressing.

Date ....
Page ....

Q2.
```
org 0000H
LJMP MAIN
ORG 0030H
MAIN:
    MOV DPTR, #250H;
    MOV R0, #40H;
    MOV R2, #06H;
LOAD_DATA:
    CLR A;
    MOVC A,@A+DPTR;
    MOV @R0,A;
    INC DPTR;
    INC R0;
    DJNZ R2, LOAD_DATA;
    MOV R0, #40H;
    MOV R2, #05H;
    MOV A, @R0;
    MOV R4, @R0;
ADD_LOOP:
    INC R0;
    ADD A,@R0;
    DA A;
    INC NEXT;
    INC R4;
NEXT:
    DJNZ R1, ADD_LOOP;
    MOV R3, A;
 H: SJMP H;
 org 250H
 DATA_1: DB 54H, 76H, 65H, 98H, 88H, 99H
 END
```

Q3.

```
Org 0000H
LJMP MAIN
Org 0030H
MAIN:
    SETB P1.0;
    MOV P0, # 0FFH;
MONITOR:
        JB P1.0, SW-HIGH;
        MOV A, P0;
        MOV P3, A;
        SETB P2.1;
        ACALL DELAY;
        SJMP MONITOR;
SW-HIGH:
        MOV A, P0;
        MOV P2, A;
        CPL P3.1;
        ACALL DELAY;
        SJMP MONITOR;
DELAY:
        MOV R6, # 250;
    D1: MOV R7, # 250;
    D2: DJNZ R7, D2;
        DJNZ R6, D1;
        RET
END
```

Q4.

	Stack Content	SP
Push 0	08H : 30H	08H
Push 1	09H : 23H	09H
Push 2	0AH : 00H	0AH
Push 1	0BH : 24H	0BH
Pop 3	24H removed	0AH
Pop 7	00H removed	09H
Push 3	0AH : 24H	0AH
Pop 6	24H removed	09H

Reg/location	value in Hex
R0	30H
R1	24H
R3	24H
R7	00h
R6	24H

Q5. (a) SETB PSW.4 ⇒ sets bits 4 of PSW (RS1=1)
MOV R2, #78H
MOV A, R2 ⇒ sets bit 3 of PSW (RS0 = 1)
SETB PSW.3
MOV R4, #0A4H
ADD A, R4
END

Since RS1=1 & RS0 = 1 Hence reg. bank 3 is selected.
→ the data 78H is stored in R2 of bank 3.
Ram address for R2 in bank 3 is 1AH.
→ the data A4H is stored in R4 of bank 3. The RAM address for R4 in Bank 3 is 1CH.

(b) 78H = 0111 1000
A4H = 1010 0100

Sum = 00011100 = 1CH with carry Out

Acc (A) = 1CH
Cy (carry flag) = 1
AC (Auxiliary Carry) = 0
OV (Overflow flag) = 1
P (parity flag) = 1

Accumulator.

(i)	ANL A, # 00H	00H
	MOV A, # 3AH	3AH
	SWAP A	A3H
	ANL A, #23H	23H
	DIV AB	08H

SLOT : D2+TD2

ANSWERS

Q1. (a) (i) error

→ can't move single bit into 8 bit accumulator

∴ MOV A, P0

(ii) error

→ decrement data pointer doesn't exist

∴ INC DPTR

(iii) error

→ can't increment an immediate value

∴ INC 40H

(iv) SETB A is not correct

→ SETB is bit addressable

∴ MOV A, #040H

(v) error

→ clr operation is bit addressable

CLR P2.0.

Date
Page

(b) $T = \frac{12}{f} = \frac{12}{15 \times 10^6 \text{ Hz}} = 0.8 \mu s$

Instruction	Machine Cycle	Executions
MOV R3, #80H	1	1
HERE: NOP	1	80
NOP	1	80
NOP	1	80
NOP	1	80
DJNZ R3, HERE	2	80
RET	2	1

⇒ Loop body : (1+1+1+1+2) x 80 = 6 X 80 = 480 cycles

Total = 1 + 480 + 2 = 483 cycles

Total Execution time = 483 X 0.8 = 386.4 μs

Q2)
```
org 0000H
START : MOV A, P1 ;
     JB ACC.7 , SQUARE_25 ;
WAVE_60 : SETB P2.1 ;
     ACALL DELAY_3 ;
     CLR P2.1 ;
     ACALL DELAY_2 ;
     SJMP START ;
SQUARE_25 :
     SETB P2.1 ;
     ACALL DELAY_1 ;
     CLR P2.1 ;
     ACALL DELAY_3 ;
     SJMP START.
```

```
DELAY-3 : MOV R0, #30;
D3      : DJNZ R0, D3;
          RET;
DELAY-2 : MOV R0, #20;
D2      : DJNZ R0, D2;
          RET;

DELAY1  : MOV R0, # 10;
D1      : DJNZ R0, D1;
          RET;
END
```

Q3.

```
ORG 0000H
MOV DPTR, #300H;
MOV R1, #10;
MOV R2, #00H;
MOV R3, #00H;
BACK : CLR A;
       MOVC A, @A+DPTR;
       ADD A, R2;
       MOV R2, A;
       JNC NEXT;
       INC R3;
NEXT : INC DPTR;
       DJNZ R1, BACK;
HERE : SJMP HERE;
       ORG 300H;
       DB 92H, 34H, 84H, 29H, 22H, 11H, 33H, 44H,
          55H, 66H;
       END.
```

Page: SP

Q4.

	data in RAM	SP
MOV 05H, #3EH;	05H : 3EH	07H
MOV 0AH, #DFH;	0AH : DFH	07H
MOV 2FH, #45H;	2FH : 45H	07H
MOV SP, #69H;	-	69H
PUSH 05H;	6AH : 3EH	6AH
PUSH 0AH;	6BH : DFH	6BH
PUSH 2FH;	6CH ; 45H	6CH
POP 4AH;	4AH : 45H	6BH
PUSH 4BH;	~~68~~ 6BH : DFH	6AH
POP 5AH;	5AH : DFH	6BH
		6AH

Q5. (i)

Instruction	Accumulator	C	RAM[20H]
MOV A, #83H	83H (1000 0011)	0	-
RLC A	06H (0000 0110)	1	-
RLC A	0DH (0000 1101)	0	-
MOV 20H, #87H	0DH	0	87H
ANL C, D2H	0DH	0	87H

Final Result :-

A : 0DH
RAM [20H] : 87H
CARRY Flag : 0

(ii)

Instruction	Register A	Register R0	RAM [40H]	Carry (C)
MOV A, #80H	80H	–	–	0
MOV R0, #40H	80H	40H	–	0
MOV @R0, A	80H	40H	80H	0
INC A	81H	40H	80H	0
RRC A	40H	40H	80H	1
ADD A, 40H	C0H	40H	80H	0

Final result :-

A : C0H
R0 : 40H
RAM [40H] : 80H
CY : 0